Karina Ponze
CPS 5745 WK9AS2

1. Skim through the topics of the accepted papers for the Vis 2026 conference: https://ieeevis.org/year/2026/program/content/.

Which topic sounds most interesting to you, so you would like to explore their paper and learn more, or, if possible, attend their presentation? Why? Justify your answer.

As I reviewed the VIS 2026 program, some topics that particularly interested me included artificial intelligence, cyber topics, and uncertainty visualization. One paper that me was "Evaluating Human-LLM Alignment on Visualization Preference," as I was curious about ways an AI-produced model might select a visualization similar to the one a human user would select. Another area of interest to me, given my background in cybersecurity, was "CUBE-ism," as I was curious about visualizations that could be useful for making decisions about which cyber events to investigate further.

The paper I'd most like to read is "A General Approach to Visualizing Uncertainty in Statistical Graphics" by Petek, Nabergoj, and Štrumbelj. In this paper, the authors sample from the data distribution several times and produce full visualizations using different samples that can be combined into a single visualization to show the overall data distribution. The author released a python package, bootplot, which I am interested in using a feature is the inclusion of separate charts showing variation between samples. I am familiar with other techniques for visualizing uncertainty, including error bars, and wanted to learn more about visualizing uncertainty in pie charts and tables.

This project reminded me of my Quantum MNIST project. On my Quantum MNIST project, one way to communicate the results of different machine learning methods was to report accuracy statistics. A potential drawback of reporting accuracy statistics is that it's hard for your audience to understand how variable the accuracy statistic is likely to be in repeated experiments. If given the chance to hear about this project at a conference or presentation, I would ask the presenters how they suggest communicating results from hybrid projects, including information about variability. The authors of this project have created a Python package that I am interested in using in repeated experiments to see if it improves communication of variability. (https://github.com/davidnabergoj/bootplot)

2. Analyze 5 attached papers of the winners of the Vis 2025 conference. What technologies/libraries/frameworks did they use? What data did they use? You can use AI for a PDF summary or Q&As. Why do you think they won?

1. **NLI4VolVis**
    - *Technology and data*. The authors used editable 3D Gaussian Splatting to build 3D scenes whose individual parts could be changed in real time. They

used SAM 2 and other methods to identify those parts, CLIP to match them to a user's words, and LLM agents to carry out the requests. The chat supports GPT-4o, DeepSeek-V3, and Llama 3.2; data was prepared in ParaView, and it ran on an NVIDIA RTX 4090 at over 100 FPS. They tested the system on eight datasets, including scans of a backpack, a carp, and human organs, as well as simulations of a hurricane, mantle temperature, and a supernova.

- *Why I think it won*. The example that stood out to me was asking to see a snake inside an egg. The system made the egg partly transparent, so the snake was visible. I can see how asking for what you want in plain language would make 3D data easier to explore, especially if you're new to the software. In the study with eight participants, about 88% of interactions used natural language.

2. **CausalMap and sentiment contagion**
    - *Technology and data*. The authors built CausalMap. They used Latent Dirichlet Allocation (LDA) to group posts into topics, an LLM to label sentiment, and a Bayesian network fitted with Fast Greedy Equivalence Search to model how sentiment might spread between topics. They tested it on two datasets, one about the NBA and basketball and one about U.S. presidential discussion. Each had more than 200,000 posts.
    - *Why I think it won*. I liked that the map lets you follow a possible path of influence over time and then open the posts behind it. Putting time on the horizontal axis also made a measurable difference. Analysts finished tasks faster and used fewer clicks and other interactions than they did with the animated version.

3. **Beyond Problem Solving**
    - *Technology and data*. The researchers interviewed 11 experienced visualization designers over Zoom and used Dovetail to transcribe and organize their notes. They looked for patterns across interviews, project diaries, sketches, and prototypes. They also gave everyone the same open-ended design challenge using the London Stage Database.
    - *Why I think it won*. This one reminded me of making charts for class. Sometimes I start with a question, make a rough chart, and realize there's a better question in the data. I think the paper stood out because the researchers used sketches and diaries to show how that change happens

while people are still designing.

4. **ReVISit 2**
    - *Technology and data*. ReVISit 2 is an open-source framework for running online studies. It uses JSON to define study steps, supports React and Vega for interactive tasks, and offers Python tools for building studies. The authors recruited 460 participants across three studies on judging correlation, designing chart textures, and searching within visualizations. They collected task responses and records of how participants interacted with the charts. Study data is saved with Google Firebase or Supabase.
    - *Why I think it won*. The replay feature stood out to me. If a participant gives an unexpected answer, I'd want to see what they clicked and what the chart looked like at that moment. Sharing the study code and settings also makes it easier for another researcher to repeat the experiment.

5. **They Aren't Built for Me**
    - *Technology and data*. The authors used swell paper, which raises printed lines and shapes when heated, to create 80 tactile bar, pie, bubble, and stacked bar charts. Eleven blind or low-vision participants estimated ratios by touch. The researchers collected their answers, response times, and comments, and four participants joined a follow-up group interview.
    - *Why I think it won*. What stuck with me was how participants used their fingers like calipers to compare lengths. Comparing the area of two bubbles took a different approach. I think the paper stood out because it showed how people actually read charts by touch and used those observations to suggest better ways to design tactile graphics.

What stood out to me is that all five papers started with a real problem people face when creating, reading, or using visualizations. They backed up their ideas with evidence from experiments, interviews, or examples from real projects. Whether the authors were making 3D data easier to explore or studying how people read charts by touch, they paid attention to how people interact with visualizations. I think that's what made these papers stand out.

3. Attempt to reproduce some of the paper's visualizations from the papers mentioned (if the code is available on GitHub). Alternatively, <u>you can choose any other paper to reproduce their Viz from Vis 2025,</u> and if you cannot obtain the PDF, I can assist with that.

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https://vc.cs.uni-magdeburg.de/assets/publications/2026/Friesecke_2026_VIS.pdf

Uncertainty-Aware PCA Revisited *Lukas Friesecke, Christian Braune, Christian Rössl, and Holger Theisel*

It is on the VIS 2025 listing.

- Locate the authors' public code repository, if available, and provide its URL.

https://github.com/lfriesecke/uncertainty-aware-pca-revisited/

- Follow the setup instructions and attempt to run the visualization.

- Record dependencies, errors, changes, and results; include an original-versus-reproduced screenshot.

Reproduction write-up

Setup. I cloned the repository and followed the README: create a virtual environment called venv_uapcar, activate it, run pip install -r requirements.txt, then run the example scripts from the src folder. The repository has two kinds of examples. The glyph examples write a 3D covariance stability glyph to an .off mesh file, and the projection examples save the sampling-based PCA projection as a PDF. I reproduced four visualizations: Fig. 5 (IRIS glyph and projection), Fig. 8 (STUDENT glyph and projection), and the right panel of Fig. 9 (ANURAN CALLS projection). My scripts, logs, and figures are in my repository.

Dependencies

Item	**Authors (README / requirements.txt)**	**My machine**
OS	Not stated	Windows 11, PowerShell
Python	3.13.5	3.10.10
numpy	2.3.1	2.2.6 (changed, see errors)
matplotlib	3.10.3	3.10.3
contourpy / pillow / others	pinned in requirements.txt	same versions
Extra (mine)	-	pymupdf (crop figures from PDF), python-docx, python-pptx

Errors and changes

Error 1: numpy 2.3.1 would not install. pip stopped with "Could not find a version that satisfies the requirement numpy==2.3.1" because numpy 2.3.x requires Python

3.11 or newer and I have Python 3.10. Change: I made requirements_py310.txt, which is the same file except numpy==2.2.6, the last release that supports 3.10. Everything else installed at the pinned versions, and the code ran without any code changes.

Issue 2: Runtime. The paper and the example scripts use U = 1,000,000 realizations, and the authors say this takes up to an hour per figure. On my laptop U = 10,000 already took about 1.3 minutes for IRIS, so 1,000,000 would have taken around 2 hours per dataset. Change: I used U = 100,000 and kept every other parameter from their examples (bins = 250, R = 0.2, the same axis limits, scale matrices, colors, and random seed). Their own parameter study (Fig. 10) shows U = 100,000 looks close to 1,000,000, with slightly noisier contour lines.

Issue 3: No glyph renderer. The glyph example only writes an .off mesh. The 3D renderer the authors used for the paper figures is not in the repository. Change: I wrote reproduce_glyphs.py, which runs their Glyph class (91 x 181 samples, as the README recommends) and draws the mesh with matplotlib using the viridis colormap and a similar camera angle.

Issue 4: Glyph axes in the wrong order. My first render did not match the paper. The IRIS glyph was stretched along z instead of lying flat. In glyph.py there is a commented-out line, "idx = np.array([2, 1, 0]) # TODO: ask if wanted", that reverses the order of the eigenvectors. The paper says the first major eigenvector should map to the z-axis, but the released code maps it to x. Reversing the eigenvector order is the same as swapping the x and z coordinates of the finished mesh, so I did that in my script and did not edit their code. After the swap, both glyphs match the paper. I kept both versions (glyph_*.png and glyph_*_swapped.png).

Minor: Matplotlib warned that C:\Users\...\.matplotlib was not writable. Setting MPLCONFIGDIR to a local folder fixed it.

Results

Figure	Script	Runtime (U = 100,000)	Match
Fig. 5 left – IRIS glyph	reproduce_glyphs.py	< 1 min	Yes, after the x/z swap
Fig. 5 right – IRIS projection	reproduce_projections.py iris	10.8 min	Yes, same clusters, shapes, and orientation
Fig. 8 left – STUDENT glyph	reproduce_glyphs.py	< 1 min	Yes, after the x/z swap
Fig. 8 right – STUDENT projection	reproduce_projections.py students	44.5 min	Yes, same layout

Fig. 9 right – ANURAN projection	reproduce_projections.py anuran	5.2 min	Yes, overall shape matches, contours a bit noisier

Overall, the reproduction worked. Every figure I tried shows the same structure as the paper. The differences are small: the contour lines are slightly wobblier because I used fewer samples, the 3D shading differs because I used a different renderer, and the glyph axes had to be swapped because of the TODO line in the code.

Original versus reproduced

Fig. 5 (left) – IRIS covariance stability glyph

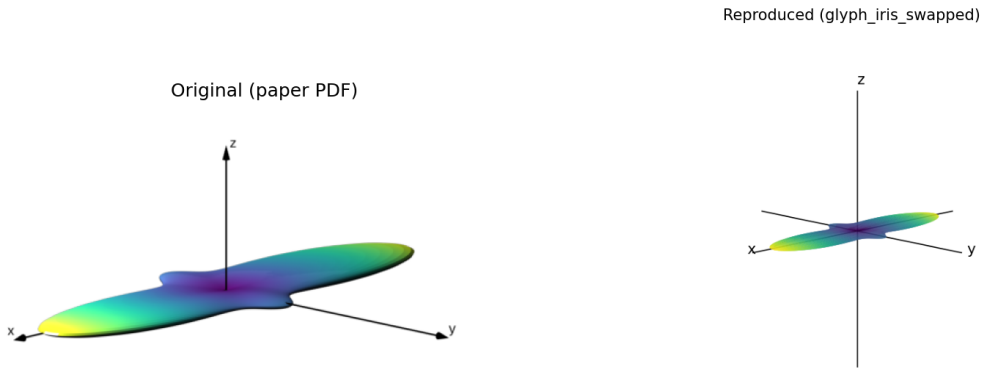


Fig. 5 (left): IRIS covariance stability glyph.

Fig. 5 (right) – IRIS sampling-based projection

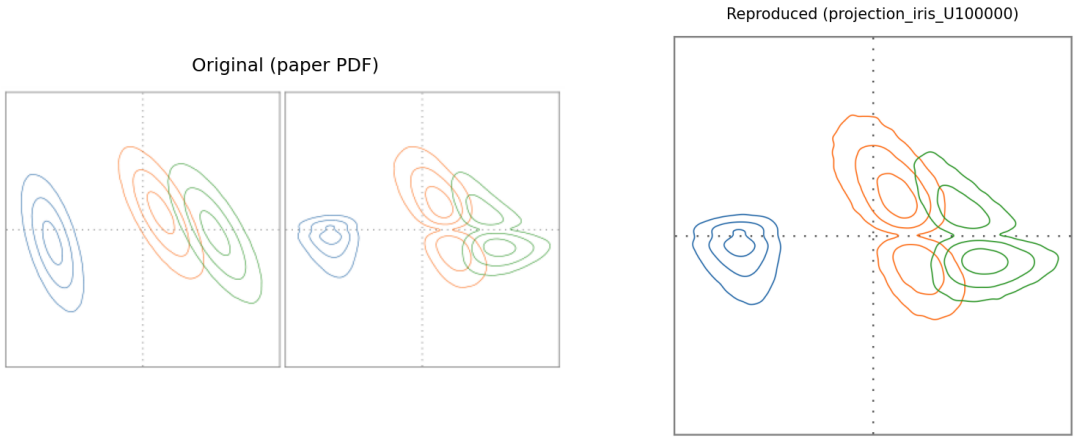


Fig. 5 (right): IRIS sampling-based projection.

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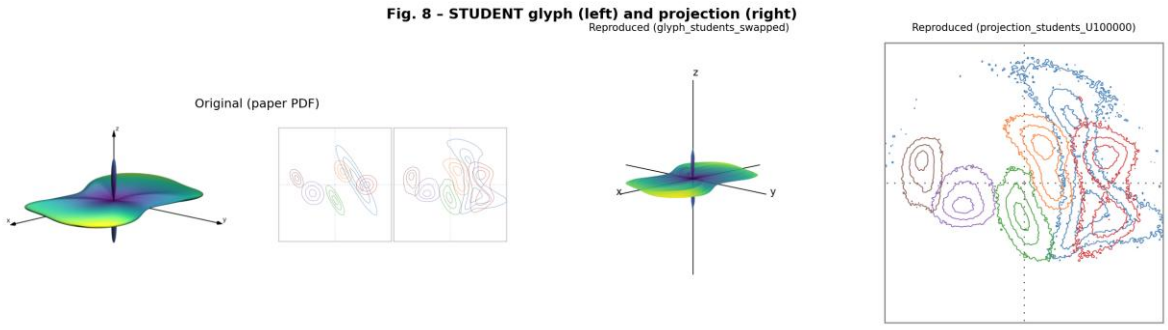


Fig. 8: STUDENT glyph and sampling-based projection.

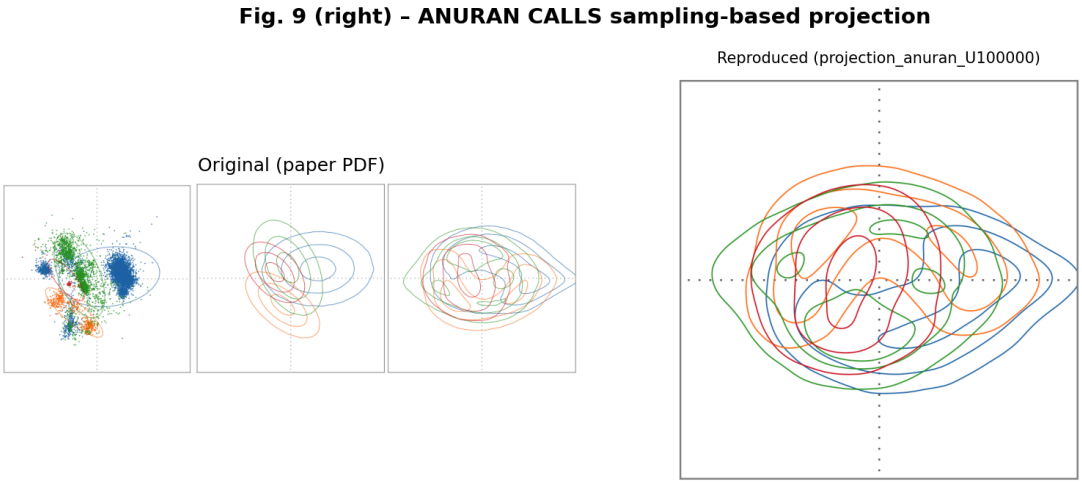


Fig. 9 (right): ANURAN CALLS sampling-based projection.

Extra credit. Mini-presentation: WK9AS2_presentation.pdf in my repository.

50 points (CSLO 1, 5)

You can obtain extra points for a mini-presentation of reproduced visualizations.